EXPERIMENT NO. 5

C++ Program: Book Class, Constructors, Destructor and Static Member

```
#include <iostream>
#include <string>
using namespace std;

class Book {
private:
    int bookId;
    string title;
    string author;
    float price;
    static int bookCount;

public:
    // Parameterized Constructor
    Book(int id = 0, string t = "Unknown",
         string a = "Unknown", float p = 0.0) {
        bookId = id;
        title = t;
        author = a;
        price = p;

        bookCount++;

        cout << "Book created. Total books : "
             << bookCount << endl;
    }

    // Copy Constructor
    Book(const Book &b) {
        bookId = b.bookId;
        title = b.title;
        author = b.author;
        price = b.price;

        bookCount++;

        cout << "Book copied. Total books : "
             << bookCount << endl;
    }

    // Destructor
    ~Book() {
        bookCount--;

        cout << "Book destroyed. Total books : "
             << bookCount << endl;
    }

    // Display Book Details
    void display() const {
        cout << "Book ID : " << bookId << endl;
        cout << "Title : " << title << endl;
        cout << "Author : " << author << endl;
        cout << "Price : " << price << endl;
    }

    // Static Function
    static int getBookCount() {
        return bookCount;
    }
};

// Initialize static member
int Book::bookCount = 0;
```

```
int main() {

    // Creating first book
    Book b1(101, "C++ Programming",
            "Bjarne Stroustrup", 4500);

    cout << "\\nBook 1 Details:" << endl;
    b1.display();

    // Creating second book using copy constructor
    Book b2(b1);

    cout << "\\nBook 2 Details:" << endl;
    b2.display();

    cout << "\\nTotal books : "
         << Book::getBookCount() << endl;

    return 0;
}
```